

UST-OLENEK, Russian Federation

WMO: 217110

Lat: 73.000N

Lon: 119.867E

Elev: 15

StdP: 101.15

Time Zone: 9.00 (E09)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-43.7	-42.5	-46.3	0.0	-43.4	-45.0	0.0	-42.3	12.4	-30.5	10.7	-31.9	4.3	140	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.9	20.5	13.8	17.9	12.3	15.2	10.7	14.5	19.3	12.6	17.2	11.2	15.0	4.2	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.1	8.8	17.4	9.9	7.6	14.4	8.3	6.8	13.3	40.8	19.6	35.6	17.5	32.2	15.0	20.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.8	8.9	DB	-44.7	23.0	2.3	3.5	-46.4	25.5	-47.7	27.5	-48.9	29.5	-50.6	32.0	(4)
(5)			WB	-44.6	15.6	2.0	1.3	-46.1	16.5	-47.3	17.3	-48.4	18.1	-49.9	19.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-12.0	-31.7	-32.4	-23.7	-14.9	-5.8	4.9	7.9	8.8	2.6	-9.1	-22.7	-29.4
	DBStd	16.16	6.64	6.63	8.23	6.11	5.05	4.86	4.53	3.48	3.51	5.78	6.77	6.23
	HDD10.0	8112	1292	1187	1044	748	491	168	97	67	222	593	982	1220
	HDD18.3	11081	1551	1420	1303	998	749	404	326	297	471	852	1232	1478
	CDD10.0	77	0	0	0	0	0	15	32	28	1	0	0	0
	CDD18.3	3	0	0	0	0	0	1	2	0	0	0	0	0
	CDH23.3	36	0	0	0	0	0	11	25	0	0	0	0	0
	CDH26.7	7	0	0	0	0	0	3	4	0	0	0	0	0
Wind	WSAvg	4.4	4.6	4.5	4.5	4.3	4.6	4.6	4.4	4.2	4.1	3.8	4.5	4.7
Precipitation	PrecAvg	307	20	12	17	14	23	21	32	45	41	38	22	23
	PrecMax	394	41	26	41	47	54	40	69	106	87	67	72	45
	PrecMin	195	4	1	4	0	5	5	11	13	13	12	4	7
	PrecStd	56	10	7	8	9	11	9	13	23	20	12	14	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-9.1	-14.8	-2.4	1.6	8.4	24.2	26.8	21.2	15.4	1.3	-2.8	-7.9
		MCWB	-9.7	-15.6	-3.2	-0.1	4.5	14.5	18.8	14.2	10.9	0.3	-3.5	-8.2
	2%	DB	-16.8	-17.6	-4.9	-1.0	5.5	19.3	21.6	19.0	10.7	0.4	-5.6	-16.0
		MCWB	-17.5	-18.2	-5.7	-2.4	2.3	12.1	15.8	12.6	7.8	-0.5	-6.4	-16.4
	5%	DB	-19.0	-20.4	-7.8	-3.1	3.0	15.6	18.4	16.5	8.7	-0.7	-10.1	-18.1
		MCWB	-19.3	-20.9	-8.6	-4.3	1.4	10.1	13.0	11.9	6.6	-1.6	-10.7	-18.5
	10%	DB	-22.5	-22.6	-11.5	-5.6	1.2	13.1	15.3	14.3	7.1	-2.1	-13.3	-20.3
		MCWB	-22.8	-22.9	-12.2	-6.6	0.0	8.8	10.6	10.6	5.3	-3.2	-13.8	-20.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-9.6	-15.3	-3.1	0.1	5.2	14.9	18.9	14.9	11.4	0.5	-3.4	-8.3
		MCDB	-9.1	-14.9	-2.4	1.3	8.1	23.7	26.6	19.3	14.6	1.1	-2.7	-7.9
	2%	WB	-17.3	-18.1	-5.7	-2.2	2.7	12.4	16.4	13.3	8.4	-0.5	-6.4	-16.3
		MCDB	-16.9	-17.8	-4.9	-1.1	5.1	18.4	21.2	17.4	10.1	0.3	-5.7	-16.0
	5%	WB	-19.3	-20.8	-8.6	-4.3	1.3	10.7	13.2	12.1	6.7	-1.6	-10.7	-18.6
		MCDB	-18.9	-20.5	-7.8	-3.1	2.8	15.3	17.2	16.2	8.4	-0.7	-10.2	-18.2
	10%	WB	-22.9	-22.9	-12.2	-6.6	0.2	8.7	11.2	10.9	5.4	-3.1	-13.8	-20.7
		MCDB	-22.6	-22.6	-11.6	-5.7	1.3	12.9	15.0	14.3	6.9	-2.2	-13.3	-20.3
Mean Daily Temperature Range	5% DB	MDBR	5.3	5.1	6.5	8.1	5.6	6.9	6.6	5.9	3.6	4.6	5.6	5.6
		MCDBR	8.7	7.9	7.7	9.2	6.7	12.9	11.9	9.6	6.8	4.4	9.1	8.3
		MCWBR	8.5	7.7	7.4	8.5	4.9	7.9	7.3	5.9	4.9	4.3	8.8	8.2
	5% WB	MCDBR	8.5	7.8	7.7	8.8	6.3	12.7	11.2	8.8	6.0	4.4	9.0	8.4
		MCWBR	8.3	7.7	7.4	8.4	5.0	8.4	7.4	6.0	4.7	4.4	8.7	8.3
Clear-Sky Solar Irradiance	taub	N/A	0.185	0.274	0.310	0.341	0.339	0.376	0.370	0.301	0.198	N/A	N/A	
	taud	N/A	1.908	2.013	2.026	2.046	2.294	2.300	2.320	2.383	1.950	N/A	N/A	
	Ebn at Noon	0	388	649	776	805	833	777	722	661	334	0	0	
	Edh at Noon	0	28	75	111	127	103	100	85	55	24	0	0	
All-Sky Solar Radiation	RadAvg	0.00	0.15	1.52	4.05	5.77	5.80	4.89	2.85	1.28	0.29	0.00	0.00	
	RadStd	0.00	0.02	0.08	0.16	0.35	0.43	0.34	0.35	0.14	0.04	0.00	0.00	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page